

APPENDIX B: Python script - syntetic data.py

```
1 | import pandas as pd
2 | import numpy as np
3 | from faker import Faker
4 | from random import randint, choice
5 | from datetime import datetime, timedelta
6 | import os
7 |
8 | fake = Faker()
9 |
10 | # =====
11 | # SETTINGS
12 | # =====
13 |
14 | NUM_CLIENTS = 100
15 | NUM_FEES = 100
16 | NUM_FACT_ROWS = 100    # keep fact rows limited to 100
17 |
18 | # =====
19 | # CREATE CLIENT DIMENSION
20 | # =====
21 |
22 | regions = ["East", "West", "North", "South"]
23 | statuses = ["Active", "Inactive"]
24 |
25 | client_rows = []
26 |
27 | for i in range(1, NUM_CLIENTS + 1):
28 |     client_rows.append({
29 |         "ClientID": f"C{i:04}",
30 |         "ClientName": fake.company(),
31 |         "Region": choice(regions),
32 |         "Status": choice(statuses)
33 |     })
34 |
35 | dim_client = pd.DataFrame(client_rows)
36 |
37 | # =====
```

```

38 | # CREATE FEE DIMENSION
39 | # =====
40 |
41 | fee_categories = [
42 |     "Subscription",
43 |     "Processing",
44 |     "Transaction",
45 |     "Adjustment",
46 |     "Penalty"
47 | ]
48 |
49 | fee_names = [
50 |     "Monthly Service Fee",
51 |     "Processing Fee",
52 |     "Late Fee",
53 |     "Usage Fee",
54 |     "Adjustment Fee",
55 |     "Transaction Fee",
56 |     "Compliance Fee",
57 |     "Premium Support Fee"
58 | ]
59 |
60 | fee_rows = []
61 |
62 | for i in range(1, NUM_FEES + 1):
63 |     fee_rows.append({
64 |         "FeeID": f"F{i:03}",
65 |         "FeeName": f"{choice(fee_names)} {i}",
66 |         "FeeCategory": choice(fee_categories)
67 |     })
68 |
69 | dim_fee = pd.DataFrame(fee_rows)
70 |
71 | # =====
72 | # CREATE FACT BILLING TABLE
73 | # =====
74 |
75 | fact_rows = []
76 |
77 | start_date = datetime(2025, 1, 1)

```

```

78 |
79 | for i in range(NUM_FACT_ROWS):
80 |
81 |     # Random client
82 |     client = dim_client.sample(1).iloc[0]
83 |
84 |     # Random fee
85 |     fee = dim_fee.sample(1).iloc[0]
86 |
87 |     # Random transaction values
88 |     units = randint(1, 25)
89 |
90 |     unit_price = round(np.random.uniform(25, 500), 2)
91 |
92 |     amount = round(units * unit_price, 2)
93 |
94 |     # Random date within 2 years
95 |     random_date = start_date + timedelta(days=randint(0, 730))
96 |
97 |     fact_rows.append({
98 |         "Date": random_date.date(),
99 |         "ClientID": client["ClientID"],
100 |         "FeeID": fee["FeeID"],
101 |         "Units": units,
102 |         "UnitPrice": unit_price,
103 |         "Amount": amount
104 |     })
105 |
106 | fact_billing = pd.DataFrame(fact_rows)
107 |
108 | # =====
109 | # EXPORT CSV FILES
110 | # =====
111 |
112 | # Create output folder if it doesn't exist
113 | output_folder = "output"
114 | os.makedirs(output_folder, exist_ok=True)
115 |
116 | dim_client.to_csv(os.path.join(output_folder, "DimClient.csv"), index=False)
117 |

```

```
118 | dim_fee.to_csv(os.path.join(output_folder, "DimFee.csv"), index=False)
119 |
120 | fact_billing.to_csv(os.path.join(output_folder, "FactBilling.csv"), index=False)
121 |
122 | # =====
123 | # PREVIEW
124 | # =====
125 |
126 | print("\nDimClient Preview")
127 | print(dim_client.head())
128 |
129 | print("\nDimFee Preview")
130 | print(dim_fee.head())
131 |
132 | print("\nFactBilling Preview")
133 | print(fact_billing.head())
134 |
135 | print("\nCSV files created successfully.")
```